GradHub

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Contents

Project Planning & Management 2

1- Project proposal 2

2- KPIs (Key performance indicators) 3

Requirements Gathering 3

1- Stakeholder Analysis 3

Primary Stakeholders 3

Secondary Stakeholders 4

External Stakeholders 4

2- User Stories & Use Cases 5

Use cases 5

User stories 6

3- Functional Requirements 8

4- Non-functional Requirements 10

Project Planning & Management

1- Project proposal

GradHub is a web platform that brings together Computer Science / IT students and HR recruiters in one place.
Instead of HR recruiters posting job ads and waiting for many unqualified applicants, they can browse student profiles directly on the website, view:

- CVs
- Graduation projects
- Skills

If a student profile matches their needs, they can open the profile and contact the student directly.
At the same time, students who are looking for jobs can:

- Create a profile
- Upload their CV
- Add their graduation project
- Browse company profiles
- See what skills companies are looking for
- Contact companies directly

Role of the Chatbot in the Project

The website includes a ready-made chatbot that helps both sides instead of manual searching.
For HR Recruiters:
HRs can type their requirements, for example:
"I am looking for a Frontend Developer with React and SQL skills."
The chatbot will suggest:

- Students whose skills match the requirements
- Or provide direct links to suitable student profiles on the platform

For Students:
Students can type their qualifications, for example:
"I am a CS graduate with .NET and SQL skills and an E-commerce graduation project."
The chatbot will suggest:

- Companies looking for similar skills
- Or suitable job opportunities

In summary, the chatbot helps:
Reduce manual searching and provides smart filtering and matching between students and companies.

2- KPIs (Key performance indicators)

1. System Performance Metrics

Response Time: The system should process search queries and filter candidates in less than 2 seconds.

System Uptime: The platform should be available 99.9% of the time (minimizing downtime during updates).

Load Handling: The system should handle at least 50 concurrent users without a decrease in performance.

2. Quality & Accuracy Metrics

AI Accuracy Rate: At least 85% of Chatbot responses should successfully guide the user to their target (Profile or Company).

Search Relevance: The "Filter by Skills" should return 100% accurate matches based on the student's database tags.

3. User Adoption & Engagement

User Adoption Rate: Aiming for 200+ active students and 20+ verified companies within the first month of deployment.

Profile Completion: At least 70% of registered students should have a "Complete" profile (CV + Project + Skills).

Recruiter Engagement: Average number of "Candidate Views" per recruiter session (Target: 10+ views).

Requirements Gathering

1- Stakeholder Analysis

Primary Stakeholders

Students

Students are one of the main users of GRADHUB. They create accounts, build professional profiles, upload CVs, showcase graduation projects, list technical skills, and apply for job opportunities. They expect an easy-to-use platform that increases their visibility to recruiters and helps them find career opportunities.

HR Recruiters

HR recruiters use the platform to search for qualified students, review profiles, download CVs, examine projects, and contact potential candidates. They expect accurate information, advanced search and filtering options, and a fast-hiring process.

Admin

The admin manages the entire platform. Responsibilities include managing users, monitoring content, resolving disputes, handling reports, maintaining security, and ensuring the system runs smoothly. The admin expects a powerful dashboard with full control over system operations.

Secondary Stakeholders

Companies

Companies benefit from the platform through their recruiters. They rely on GRADHUB to reduce hiring time, discover skilled graduates, and improve recruitment efficiency.

Universities

Universities benefit when their students gain internships and jobs. The platform can improve graduate employability and strengthen the university's reputation.

Project Supervisors/ Academic Staff

Supervisors may verify student projects or achievements if such a feature is added. They are interested in ensuring student work is represented accurately.

Support Team

The support or maintenance team handles technical problems, user issues, updates, and bug fixes. They need clear system logs, maintainable code, and support tools.

External Stakeholders

Hosting / Cloud Provider

The hosting provider supplies the infrastructure that keeps GRADHUB online, secure, and scalable.

Government / Regulatory Authorities

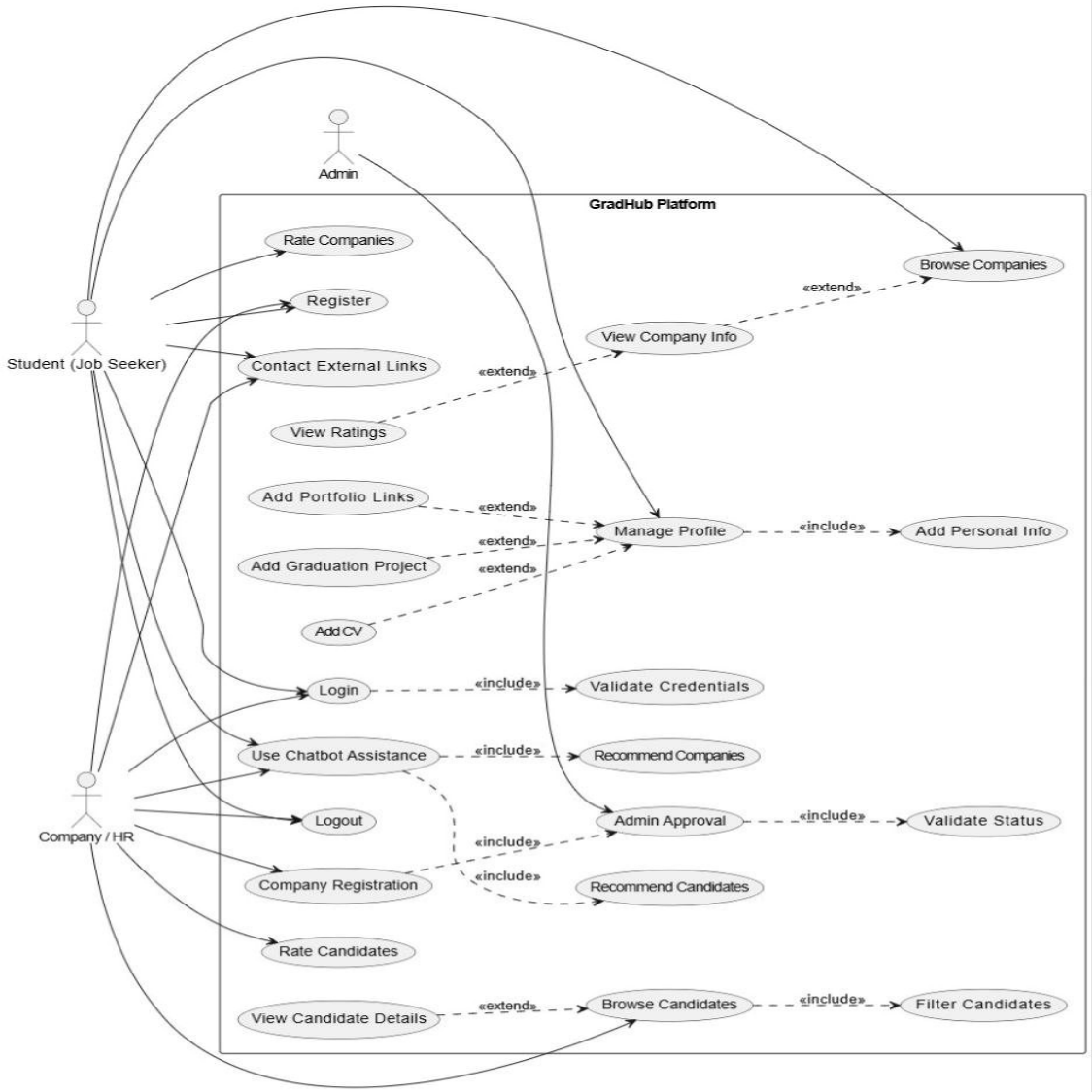
Authorities may require compliance with privacy, cybersecurity, and data protection laws.

Third-Party Services

External services such as email providers or authentication tools may be used for verification of emails, notifications, or login integration.

2- User Stories & Use Cases

Use cases



User stories

1. Student Actor

User Story: Profile Management
As a student I want to create and manage my professional profile so that I can showcase my skills and projects to potential recruiters.
Acceptance criteria - The system shall allow the user to edit personal information (Name, Field, Bio, Skills). - Name and Field of Study fields must be mandatory - The CV upload feature must only accept PDF files (Max 5MB) - Portfolio links, GitHub, LinkedIn must undergo URL format validation. - The graduation project section must allow for a title, description, and link

User Story: Company Discovery
As a student I want to browse registered companies and see their ratings so that I can apply to reputable workplaces
Acceptance criteria - The system must display a list of all verified companies. - Each company card must show the average star rating (1–5). - Clicking on a company must display its full details and reviews

User Story: AI Chatbot Assistance
As a student I want to use a chatbot to suggest companies so that I can find opportunities that match my skills without manual searching
Acceptance criteria - The chatbot must recognize field-related keywords (e.g., ".NET," "Frontend"). - The bot must provide direct links to the "View Company Info" page. - The system must provide a fallback response if the query is not understood.

2. Recruiter Actor

User Story: Candidate Search & Discovery
As a Recruiter I want to search and filter through student candidates so that I can efficiently find the right talent for my team
Acceptance criteria - The search bar must support queries by Name or Field. - The system must provide filters for technical skills - Search results must display candidate cards with a summary of skills and ratings. - Clicking on a card must show the student's full profile.

User Story: Candidate Evaluation
As a Recruiter I want to rate candidates after reviewing them so that I can track high-quality leads.
Acceptance criteria - The system shall allow a star rating (1–5) and an optional text review. - A recruiter cannot rate the same candidate more than once. - Ratings must be editable only by the recruiter who posted them.

User Story: Smart Recruitment (Chatbot)
As a Recruiter I want to ask the Chatbot to find specific candidates so that I can speed up my recruitment process.
Acceptance criteria - The chatbot must respond to queries like "Find developers with SQL skills." - The chatbot must return a filtered list of candidate profiles.

3. Admin Actor

User Story: Company Verification
As an admin I want to review and approve company registration requests so that the platform remains secure
Acceptance criteria - The dashboard must list all companies with "Pending" status. - The admin must be able to view verification documents uploaded by the company. - Upon approval, the account status must be updated to "Approved" (Include: Update Account Status). -An automated notification must be triggered by the company representative upon a status change.

3- Functional Requirements

1. **User Registration and Authentication**
   - The system shall allow users to register as Job Seekers (Students) or Companies.
   - The system shall allow users to log in and log out securely.
   - The system shall validate user credentials during login.

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2. **Job Seeker Profile Management**
   - The system shall allow job seekers to create and edit their profiles.
   - The system shall allow job seekers to add personal information such as name, field, and skills.
   - The system shall allow job seekers to upload their CV.
   - The system shall allow job seekers to add portfolio links such as LinkedIn, GitHub, or Behance.
   - The system shall allow job seekers to update their profile information at any time.

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3. **Company Registration and Verification**
   - The system shall allow companies to create company accounts.
   - The system shall require admin approval before a company can access candidate profiles.
   - The system shall store the verification status of each company (Pending / Approved / Rejected).

4. **Candidate Browsing**
   - The system shall allow verified companies to browse job seeker profiles.
   - The system shall allow companies to view candidate details including skills, CV, and portfolio links.
   - The system shall allow companies to search and filter candidates based on skills or field.

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5. **Company Browsing**
   - The system shall allow job seekers to browse registered companies.
   - The system shall allow job seekers to view company information and ratings.

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6. **External Communication**
   - The system shall allow companies to contact candidates through external communication links such as LinkedIn, WhatsApp, or Email.
   - The system shall allow job seekers to contact companies through the same external channels.

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7. **Chatbot Assistance**
   - The system shall integrate a chatbot to assist recruiters in filtering candidates based on required skills.
   - The system shall allow job seekers to use the chatbot to discover companies related to their skills.

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8. **Rating System**
   - The system shall allow companies to rate candidates after interaction.
   - The system shall allow candidates to rate companies.
   - The system shall display ratings on user profiles to increase transparency.

4- Non-functional Requirements

1. **Performance Requirements**
   - The system shall load main pages within 3 seconds under normal conditions.
   - Search results shall be displayed within 2 seconds.
   - The platform shall support multiple users at the same time without major slowdown.
   - File uploads such as CVs and project documents shall be completed efficiently depending on file size and internet speed.

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2. **Security Requirements**

- Users shall log in using secure authentication.
- Passwords shall be encrypted or hashed before storage.
- Only authorized users may access restricted features.
- Students cannot access admin functions.
- Recruiters cannot modify student profiles.

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3. **Usability Requirements**

- The interface shall be simple, clean, and easy to navigate.
- Error messages should be understandable and helpful.
- The system shall be responsive on desktop, tablet, and mobile devices.

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4. **Reliability Requirements**
   - The system shall operate consistently with minimal failures.
   - Uploaded files and stored data shall not be lost during unexpected errors.

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5. **Availability Requirements**
   - The platform shall be available 24/7 except during scheduled maintenance.
   - Planned maintenance should be announced in advance when possible.

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6. **Scalability Requirements**
   - The system shall support growth in the number of students, recruiters, and companies.
   - Database performance shall remain acceptable as data increases.
   - New modules or services should be added without redesigning the whole system.

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7. **Maintainability Requirements**
   - The code shall be modular and well-structured.
   - Documentation shall be available for future developers.
   - Updates and new features should be implemented with minimal disruption.